

Demystifying Agentic AI

Tool Calling, Skills, and Modern Architecture

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The task

“The test suite is failing. Fix it.”

What `run_tests()` shows (excerpt):

```
def test_median_even():
>     assert median([4, 1, 3, 2]) == 2.5
E     assert 3 == 2.5
E         + where 3 = median([4, 1, 3, 2])

FAILED test_stats.py::test_median_even
1 failed, 2 passed in 0.01s
```

The code under test, `stats.py`:

```
def mean(xs):
    return sum(xs) / len(xs)

def median(xs):
    xs = sorted(xs)
    n = len(xs)
    return xs[n // 2]
```

A coding agent at work

> The test suite is failing. Fix it.

- I'll start by running the tests.

- `run_tests()`

```
FAILED test_stats.py::test_median_even
```

```
assert 3 == 2.5 (median([4, 1, 3, 2]) is 3)
```

```
1 failed, 2 passed
```

A coding agent at work

✓ `run_tests()` 1 failed, 2 passed

- The failure is in median. Let me read it.

- `read_file("stats.py")`

```
def median(xs):  
    xs = sorted(xs)  
    n = len(xs)  
    return xs[n // 2]
```

A coding agent at work

✓ `run_tests()` 1 failed, 2 passed

✓ `read_file("stats.py")`

- For even lengths it should average the middle two.

• `edit_file(`

`path: "stats.py",`

`old_text: "return xs[n//2]",`

`new_text: "if n % 2 == 0:`

`return (xs[n // 2 - 1] + xs[n // 2]) / 2`

`return xs[n // 2]"`)

`error: old_text not found in stats.py`

A coding agent at work

```
✓ run_tests() 1 failed, 2 passed
✓ read_file("stats.py")
✗ edit_file(...) error: old_text not found
```

- The spacing didn't match. Retrying with the exact text.
- `edit_file(`
 `path: "stats.py",`
 `old_text: "return xs[n // 2]",`
 `new_text: "if n % 2 == 0: ... return xs[n // 2]"`
 `ok`

A coding agent at work

```
✓ run_tests() 1 failed, 2 passed
✓ read_file("stats.py")
✗ edit_file(...) error: old_text not found
✓ edit_file(...) ok

● run_tests()
  3 passed

● Fixed: median now averages the two middle values for
  even-length lists. All 3 tests pass.
```

That looks like magic. What is going on?

An LLM

reads the history
and chooses
the next action

Three tools

read_file,
run_tests, edit_file

A loop

runs the action,
adds the result to
the history, repeats

Next: each piece, with the math that pins it down.

What is a tool?

$$\tau = (\text{name}, S, f), \quad S \subseteq \mathcal{X}, \quad f : S \rightarrow \mathcal{Y}$$

Our example, $\tau = \text{edit_file}$:

$\mathcal{X}$ all argument objects: key–value pairs of strings (JSON)

S objects with exactly the keys `path`, `old_text`, `new_text`

f replace `old_text` by `new_text` in the file; return “ok” or an error message

$\mathcal{Y}$ text

The model sees (name, S) and writes down a call. The host holds f : it checks $x \in S$, then runs f . If $x \notin S$, it returns an error message instead.

Tools in the example

name	S : allowed arguments	f : what the host does	result in $\mathcal{Y}$
read_file	a path (string)	reads the file	its text
run_tests	none, or a path	runs the test suite	pass/fail summary, traceback
edit_file	a path, the old text, the new text	replaces old with new	“ok” or an error message

S checks only the *shape* of the arguments. Whether `old_text` occurs in the file is checked when f runs, and a failure comes back as an ordinary result in $\mathcal{Y}$.

$T = \{\tau_1, \dots, \tau_k\}$ denotes the agent's set of tools.

What the LLM sees and can do

Sees a history $h_t \in H$: the user's request, then one (action, result) pair per step so far.

$h_2 = \left(\text{"The test suite is failing. Fix it."}, \right.$
 $\left(\text{call}(\text{run_tests}, \{\}), \text{"1 failed, 2 passed"} \right),$
 $\left. \left(\text{call}(\text{read_file}, \{\text{path: "stats.py"}\}), \text{"def median(xs): ..."} \right) \right)$

Can choose an action $a_t \in A = \{\text{answer}(m)\} \cup \{\text{call}(\tau, x) : \tau \in T, x \in \mathcal{X}\}$:

<code>call(edit_file, {path: "stats.py", ...})</code>	run a tool τ on arguments x
<code>call(run_tests, {})</code>	run a tool τ on arguments x
<code>answer("Fixed median; 3 tests pass.")</code>	reply to the user with text m , and stop

H is the set of all such histories; x can be any argument object in $\mathcal{X}$.

The LLM chooses

$$\pi : H \rightarrow \Delta(A), \quad a_t \sim \pi(\cdot \mid h_t)$$

$\Delta(A)$ is the set of probability distributions over A . Given the history, the LLM puts a probability on every possible next action, and one is drawn at random ($\sim$).

$\pi(\cdot \mid h_2)$ for our example (made-up numbers)

`call(edit_file, {...})`  0.80

`call(run_tests, {})`  0.10

`call(read_file, {...})`  0.07

everything else, incl. `answer(m)`  0.03

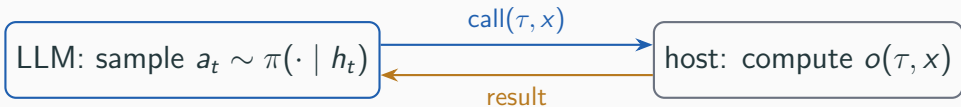
How π learns to choose well comes from training; for this talk it is a black box we sample from.

The loop

$$o(\tau, x) = \begin{cases} f_\tau(x) & x \in S_\tau \\ \text{an error message} & \text{otherwise} \end{cases}$$

$$h_{t+1} = h_t \frown (a_t, o(\tau, x)) \quad \text{if } a_t = \text{call}(\tau, x)$$

stop when $a_t = \text{answer}(m)$ or $t = N$



$\frown$ appends one entry to the end of the history. h_0 is the user's request. N is a step budget.

The loop, on our example

h_0 = "The test suite is failing. Fix it." Each step appends one (action, result) pair.

t	action a_t	result $o(\tau, x)$
0	call(run_tests, {})	1 failed, 2 passed
1	call(read_file, {path: "stats.py"})	the source of stats.py
2	call(edit_file, {old_text: "return xs[n//2]", ...})	error: old_text not found
3	call(edit_file, {old_text: "return xs[n // 2]", ...})	ok
4	call(run_tests, {})	3 passed
5	answer("Fixed median; 3 tests pass.")	(loop stops)

Be the policy (2.5 min)

The history h_2 for a new bug:

```
> Fix the failing tests.  
run_tests()  
  FAILED test_one, test_zero  
  where True = is_prime(1)  
  where True = is_prime(0)  
  2 failed, 2 passed  
read_file("numtheory.py")  
def is_prime(n):  
    for d in range(2, int(n ** 0.5) + 1):  
        if n % d == 0:  
            return False  
    return True
```

What should a_2 be? **Vote.**

- (a) `call(run_tests, {})`
- (b) `call(edit_file, {path:
"numtheory.py", ...})`
- (c) `call(edit_file, {path:
"test_numtheory.py",
...})`
- (d) `answer("The tests are
wrong.")`

Be the policy (2.5 min)

The history h_2 for a new bug:

```
> Fix the failing tests.  
run_tests()  
FAILED test_one, test_zero  
where True = is_prime(1)  
where True = is_prime(0)  
2 failed, 2 passed  
read_file("numtheory.py")  
def is_prime(n):  
    for d in range(2, int(n ** 0.5) + 1):  
        if n % d == 0:  
            return False  
    return True
```

What should a_2 be? **Vote.**

- (a) `call(run_tests, {})`
repeats what we know; wastes a step
- (b) `call(edit_file, {path:
"numtheory.py", ...})`
reasonable: adds the missing $n < 2$ case
- (c) `call(edit_file, {path:
"test_numtheory.py",
...})`
passes without fixing the code
- (d) `answer("The tests are
wrong.")`
not supported by the history

The room's votes are an estimate of $\pi(\cdot \mid h_2)$.

Be the host (2.5 min)

You are the host. The tool is $\tau = \text{edit_file}$, with $S =$ objects with keys `path`, `old_text`, `new_text`, all strings. The file is `numtheory.py` from the last slide. For each call, what is $o(\tau, x)$?

call

- 1 `old_text: "for d in range(2, int(n ** 0.5) + 1):",`
`new_text: "..."`
- 2 `old_text: "for d in range(2, int(n**0.5) + 1):",`
`new_text: "..."`
- 3 `old_text: "return True"`
(no `new_text` key)
- 4 `old_text: "return",`
`new_text: "yield"`

In every call, `path` is `"numtheory.py"`.

Be the host (2.5 min)

You are the host. The tool is $\tau = \text{edit_file}$, with $S =$ objects with keys `path`, `old_text`, `new_text`, all strings. The file is `numtheory.py` from the last slide. For each call, what is $o(\tau, x)$?

call	$o(\tau, x)$
1 <code>old_text: "for d in range(2, int(n ** 0.5) + 1):",</code> <code>new_text: "..."</code>	ok
2 <code>old_text: "for d in range(2, int(n**0.5) + 1):",</code> <code>new_text: "..."</code>	error: <code>old_text</code> not found in <code>numtheory.py</code>
3 <code>old_text: "return True"</code> (no <code>new_text</code> key)	error: invalid arguments for <code>edit_file</code> (expected keys: <code>path</code> , <code>old_text</code> , <code>new_text</code>)
4 <code>old_text: "return",</code> <code>new_text: "yield"</code>	error: <code>old_text</code> occurs 2 times in <code>numtheory.py</code>

In every call, `path` is `"numtheory.py"`.

Debrief (1 min)

1. Where did the *choosing* happen, and where did the *checking* happen?

the policy π versus the schema S and the function f

2. Call 3 was rejected before f ran. Why is it still useful that the model sees that error?

errors are observations, so the loop can recover

3. If `edit_file` could also delete files, what would you change?

a scoped tool set $T_t \subseteq T$

What is a skill?

A folder of instructions, plus optional scripts. It runs nothing by itself.

```
fix-failing-tests/  
  SKILL.md
```

```
---  
name: fix-failing-tests  
description: Use when asked to fix  
  failing tests or a failing build.  
---  
1. Run the full test suite first; read  
  the failure before opening any file.  
2. Read the code under test, not just  
  the test.  
3. Make the smallest change that fixes  
  the failure.  
4. Do not edit a test to make it pass  
  unless asked.  
5. Rerun the full suite, not just the  
  failing test.  
6. Report what you changed and why.
```

How the agent uses a skill

- **Session start:** the agent sees only each skill's name and one-line description.

How the agent uses a skill

- **Session start:** the agent sees only each skill's name and one-line description.
- **A task arrives** (“fix the failing tests”): the description matches, so the agent reads `SKILL.md` with an ordinary file-read call.
- **Now** the instructions are in its context. It may also run the bundled script, another ordinary tool call.

A tool is something the agent can *do*. A skill is something it *knows how to do well*.

Why load on demand

Context budget. n skills, each with a description of d tokens and instructions of c tokens; k skills used:

$$\underbrace{n(d + c)}_{\text{load everything}} \quad \text{vs.} \quad \underbrace{nd + kc}_{\text{index, then load on demand}}$$

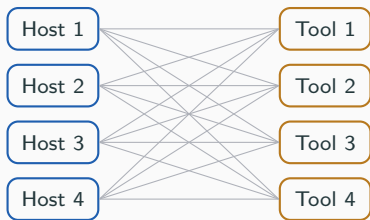
Illustrative numbers: $n = 50$, $d = 50$, $c = 2000$, $k = 1$ gives 102,500 versus 4,500 tokens.

Loading a skill is a tool call. It changes what the model conditions on, not the model itself:

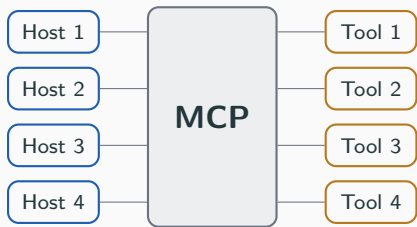
$$h_{t+1} = h_t \curvearrowright (\text{call}(\text{read_file}, \text{SKILL.md}), \text{ text of SKILL.md})$$

Scoping. At each step the agent sees only $T_t \subseteq T$: fewer ways to be wrong, and least privilege.

MCP: one protocol instead of $N \cdot M$ integrations



$N \cdot M = 16$ integrations



$N + M = 8$ implementations

Each AI application implements one side of a shared protocol once, and each tool implements the other side once. For $N = 5$ and $M = 8$: $40 \rightarrow 13$.

Takeaways

1. An agent is a model in a loop: it emits structured calls, the host validates and runs them, and the results go back into the history.
2. Reliability is a design problem: precise schemas, small scoped tool sets, and verification steps (like rerunning the tests) make errors easier to catch.
3. Skills and MCP are architecture, not magic: skills load know-how on demand, and a shared protocol turns $N \cdot M$ integrations into $N + M$.

Questions?

[TODO: add links (MCP specification, exercise handout) and contact info]

Backup: the median bug and its fix

before

```
def median(xs):  
    xs = sorted(xs)  
    n = len(xs)  
    return xs[n // 2]
```

after

```
def median(xs):  
    xs = sorted(xs)  
    n = len(xs)  
    if n % 2 == 0:  
        return (xs[n // 2 - 1]  
                + xs[n // 2]) / 2  
    return xs[n // 2]
```

Backup: exercise answer key

Fix: add the $n < 2$ case. All 4 tests pass afterward.

```
def is_prime(n):  
    if n < 2:  
        return False  
    for d in range(2, int(n ** 0.5) + 1):  
        if n % d == 0:  
            return False  
    return True
```